Declining silicate and nitrate concentrations in the northern North Atlantic

Kjell Gundersen, Vidar S. Lien, Jane S. Møgster, Jan Even Øie Nilsen, Håvard Vindenes (IMR)

Corresponding author: *Vidar S. Lien (vidar.lien@hi.no)*

Statement of main outcome

A comprehensive analysis of nutrient data in three regions of the Nordic Seas between 1990 and 2019, shows a statistically significant decline in surface silicate. This finding is in agreement with previous reports on silicate from the northern regions of the North Atlantic. However, this is the first look at a 30-year record of water column silicate and nitrate in the Nordic Seas. Surface nutrient concentrations in the Nordic Seas appears to be regulated by the Subpolar Gyre situated south of Greenland and Iceland. The Subpolar Gyre Index has been in a decline for most of the period investigated, which means that the Subpolar Gyre has moved westward allowing more subtropical (and more nutrient depleted) water into the Nordic Seas. The largest decline in silicate occurred during the first ten years investigated and is still on a downward slope. We also found a statistically significant decline in Arctic Water in the Greenland Sea, but with a time-lag relative to the decline in Atlantic Water. Nitrate on the other hand, did not decline as uniformly and only had a significant drop midway through the time-series (2005-2009) and only in the Norwegian Sea, before it again increased to previous levels. The molar nitrate:silicate-ratio however, showed a steady increase throughout the thirty year period investigated. Less access to silicate and other macronutrients in the Nordic Seas may shorten the spring diatom bloom period and hamper zooplankton growth, which in turn may have consequences for growth and development of commercially important fish stocks in these waters.

Name of expert	Affiliation	Full address	Email
Kjell Gundersen	Institute of Marine Research (IMR)	Nordnesgaten 50, 5005 Bergen, Norway	kjell.gundersen@hi.no
Vidar S. Lien	Institute of Marine Research (IMR)	Nordnesgaten 50, 5005 Bergen, Norway	vidar.lien@hi.no
Jane S. Møgster	Institute of Marine Research (IMR)	Nordnesgaten 50, 5005 Bergen, Norway	jane.stroemstad@hi.no
Jan Even Øie Nilsen	Institute of Marine Research (IMR)	Nordnesgaten 50, 5005 Bergen, Norway	jan.even.oie.nilsen@hi.no
Håvard Vindenes	Institute of Marine Research (IMR)	Nordnesgaten 50, 5005 Bergen, Norway	haavard.vindenes@hi.no

1.4.1. Introduction

Periodic changes in the eastward extent of the Subpolar Gyre (SPG), an important regulator of the North Atlantic thermohaline circulation, determines the inflow of high-saline water to the Nordic Seas (Hatun et al. 2005). Rey (2012) first noticed a bidecadal drop in dissolved silicic acid (silicate) concentrations in the Nordic Seas, along with an increase in salinities, and suggested that periods of relative weak North Atlantic Oscillation (NAO) caused warmer, high-saline Atlantic Water (AW) to enter the Norwegian Sea. The strength of the NAO determines the east-west positioning and frontal shifts of the SPG and hence, the magnitude of AW influx to the Norwegian Sea (Sarafanov 2009). More recently, Hatun et al. (2017) were able to demonstrate that the shifting fronts of the SPG to a large extent determine surface hydrography, and silicate content, in several areas of the subpolar North Atlantic region (Norwegian Sea, Irminger Sea, Labrador Sea).

Extended periods of reduced access to silicate, and possibly other nutrient elements, may have severe implications for the pelagic community composition in the Nordic Seas. North Atlantic spring blooms are often dominated by diatoms (Savidge et al. 1995) as the primary consumers of dissolved silicate rapidly respond to the high nutrient concentrations. Toward the end of the bloom however, silicate and other macronutrients are approaching depletion and the phytoplankton community becomes less buoyant. As the remnants of the spring bloom exit surface waters it becomes a major component of the export of particulate organic matter to the deep ocean (Honjo & Manganini 1993). The latter has led Pollock (1997) and others to suggest that available silicate in surface waters may determine atmospheric CO_2 drawdown via the "biological pump" (Volk & Hoffert 1985). However, an overall decline in accessible silicate in surface waters, the end product of deep winter mixing events with less silicate rich deep water, may reduce the length and magnitude of the annual phytoplankton spring bloom.

Diatoms are only competitive with other phytoplankton species at silicate concentrations >2 µM (Furnas 1990, Egge & Aksnes 1992). Therefore, reduced annual influx of silicate to the euphotic zone, relative to other macronutrients, may hasten the switch towards a flagellate dominated community and regenerated production (Smayda 1990). As of today, spring blooms in the Nordic Seas are mainly composed of diatoms (Rey 2004) and the magnitude and timing of each bloom is considered a deciding factor in both growth and recruitment of *Calanus finmarchicus* and *C. glacialis*, two of the most important calanoid copepods in the region (Melle et al. 2004). Calanoid copepods, such as the most abundant species, *C. finmarchicus*, appears to have a preference for diatoms (Meyer-Harms et al. 1999) and a number of studies have concluded that food quality affects copepod production and ultimately, growth at higher trophic levels (Barofsky et al. 2010, and references therein). Calanoid copepods in turn, are considered one of the main staples for several species of commercially important pelagic fish, both in the Norwegian Sea and the Barents Sea (Skjoldal et al. 2004, Gjøsæther 2009). Therefore, changes in the magnitude or composition of the diatom spring bloom may ultimately have direct commercial implications for commercial fisheries in these regions.

Here we present an updated, 30 year time-series of silicate measurements from the Nordic Seas, as we also consider the fate of nitrate concentrations in these waters during the same period of time. The relative change between the two macronutrient concentrations (the nitrate:silicate-ratio) and its implications for phytoplankton growth and community composition, is also discussed.

1.4.2. Data and Methods

Seawater samples were collected from Niskin-type water bottles at predetermined depths triggered by a Conductivity Temperature Depth (CTD) instrument package mounted on a rosette. Up until the last millenium, the majority of nutrient samples were analyzed in real time onboard the ships. As the research fleet expanded at IMR, the number of autoanalyzers could no longer match the number of ships operating simultaneously, and nutrient samples (20 mL) were added chloroform (200 µL) and stored in the fridge at +4 ºC for analysis at the home laboratory within 1-6 weeks after collection. The samples were allowed to acclimate to room temperature as the chloroform was evacuated by vacuum, prior to analysis on an Automated Analyzer (AA) system. With one exception, all nutrients were run on homemade AA system assemblies (*Skalar* and *Alpkem* hybrids) up until recently. The latest upgrade was the first complete AA system purchased from *Skalar Analytical B.V.* (Breda, The Netherlands). Colometric determinations of dissolved inorganic nutrients are based on the methods first described by Bendschneider & Robinson (1952) and Grasshof (1965) with a number of minor adjustments suggested by the manufacturers (*Alpkem*, *Skalar Analytical*). The AA system measures nitrate (NO_3^-), nitrite (NO_2^-), phosphate (PO_4^-) and silicate (SiO_4^-) but only nitrate and silicate are reported in this study. Briefly, nitrate in seawater is reduced to nitrite coupled to a diazonium ion and, in the presence of aromatic amines, the resulting blue azo-dye is determined spectrophotometrically at 540 nm. The nitrate concentration is corrected for ambient nitrite (same analytical method as for nitrate, but without cadmium reduction) measured concurrently. Silicate (silicic acid) is reacting to molybdate at low pH and the resulting silicomolybdate is reduced by ascorbic acid to a blue dye measured spectrophotometrically at 810 nm.

The Plankton Chemistry Laboratory at IMR maintains quality control of precision and accuracy by daily assessments of analytical standard curves and internal standards. Both nitrate and silicate concentrations are measured with a precision < 0.2 % and the accuracy deviated < 1 % from the internal standard. The laboratory is an active participant in the biannual intercomparisons initiated by the *QUASIMEME Laboratory Performance Studies* (http://www.quasimeme.org/), as well as other inter-laboratory comparisons, such as the *International Ocean Carbon Coordination Project* (IOCCP) intercalibration of nutrient analysis in 2017 (https://repository.oceanbestpractices.org/handle/11329/883).

This data set contains nutrient measurements, including silicate and nitrate, collected from the Nordic Seas between 1990 and 2019 (Fig. 1.4.1). All data were quality controlled by the analysts using quality flags 1-5 in accordance with Jaccard et al. (2018) and incorporated into the Copernicus Marine Environment Monitoring Service (CMEMS) *in situ* database (product ref. 1.4.1). All data in the CMEMS *in situ* database are further scrutinized through automated tests and all outliers are checked manually (Jaccard et al. 2020). Only data flagged as “good data” (flag = 1) were included in this study.

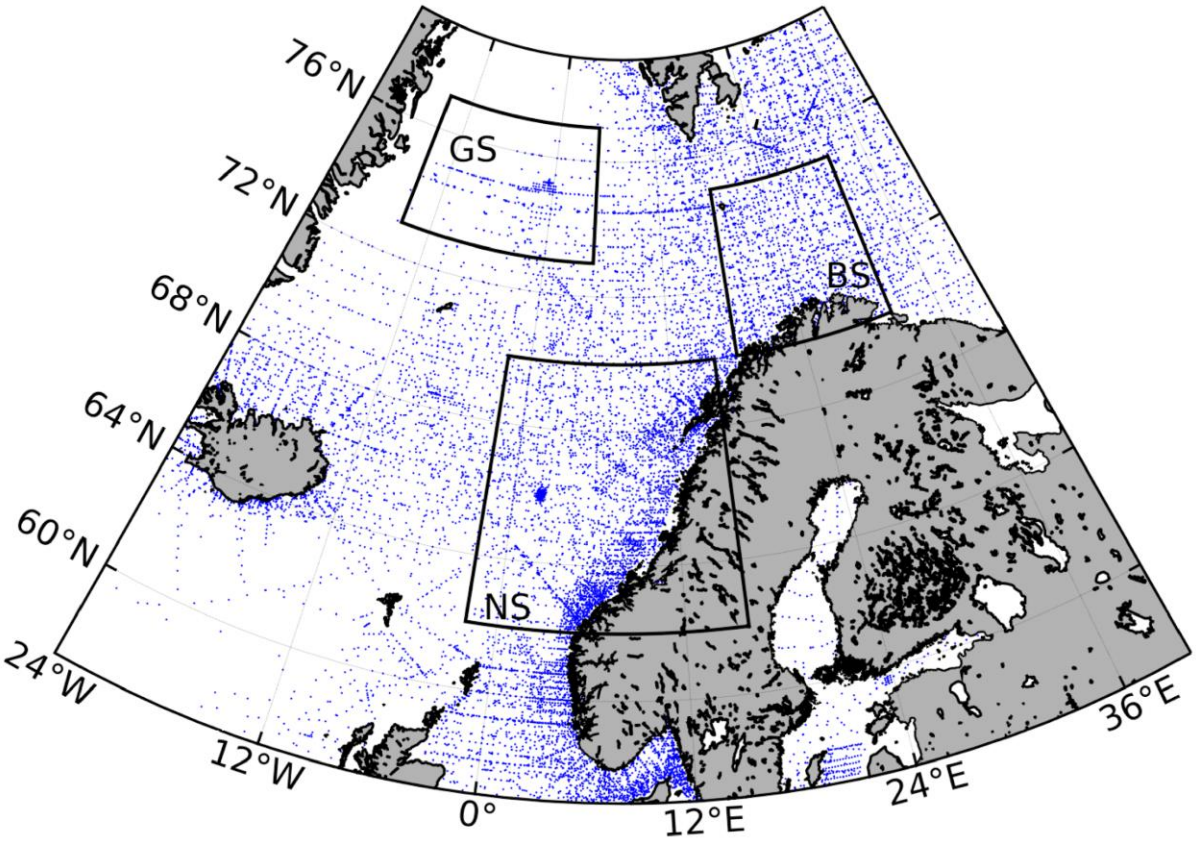


Figure 1.4.1. *The Nordic Seas and all stations visited (blue dots) where nutrient data have been collected (product ref. 1.4.1) and analysed in the period investigated (1993-2016). Black rectangles show the regions with data used in this study: NS (Norwegian Sea), BS (Barents Sea), GS (Greenland Sea).*

In this study we focus on the deep water reservoir of nutrients in order to further document the decadal decline already reported in literature (Rey 2012, Hatun et al. 2017). Because the main source of nutrients in the Nordic Seas is the North Atlantic, we used silicate and nitrate data from high-saline Atlantic Water collected in winter (January-March) in order to avoid effects of biological uptake by primary production. Therefore, we limit our study to silicate and nitrate data from three regions (Fig. 1.4.1) and only in AW (here defined by $S > 34.9$) in order to avoid continued supply of nutrients to surface waters through the year, as is the case with river runoff in coastal waters. In the Greenland Sea, AW influence is mainly from the subsurface Return Atlantic Water from the Fram Strait in the north. We divided the data into two depth segments; the euphotic zone (0-50 m), and the core of Atlantic Water flow through the Nordic Seas below the euphotic zone (100-200 m). Due to poor cruise coverage in winter in the Greenland Sea, we used deep water data (only 100-200 m range) from the whole year in that region.

1.4.3. Results

Surface and deep water silicate concentrations from the three selected regions were binned in 5-year periods (Fig. 1.4.2). No available data from the Barents Sea in the second period (1995-1999) met our selection criteria (region and salinity range) and is therefore not shown. There is a significant drop in

silicate concentrations from the first two periods (1990-1999) to the subsequent periods in the Norwegian Sea, while in the Barents Sea the drop continues through the first half of year 2000 (Fig. 1.4.2). The largest decrease appeared in deeper layer (100-200 m) in the Barents Sea, where silicate concentrations dropped by 19 % ($p < 0.05$) from 1990-1994 to 2015-2019. A statistically significant ($p < 0.05$) decrease of comparable magnitude (17 %) was also observed in the euphotic zone (0-50 m) between the same periods. The decline in the Norwegian Sea, during the same two periods of time, were 17 % in surface waters and 16 % in deeper waters. A larger part of the decrease occurred during the 1990-2004 period, followed by an apparent levelling off after 2005. The last period (2015-2019) showed a statistically significant decline compared to 2010-2014 in the Barents Sea (Fig. 1.4.2).

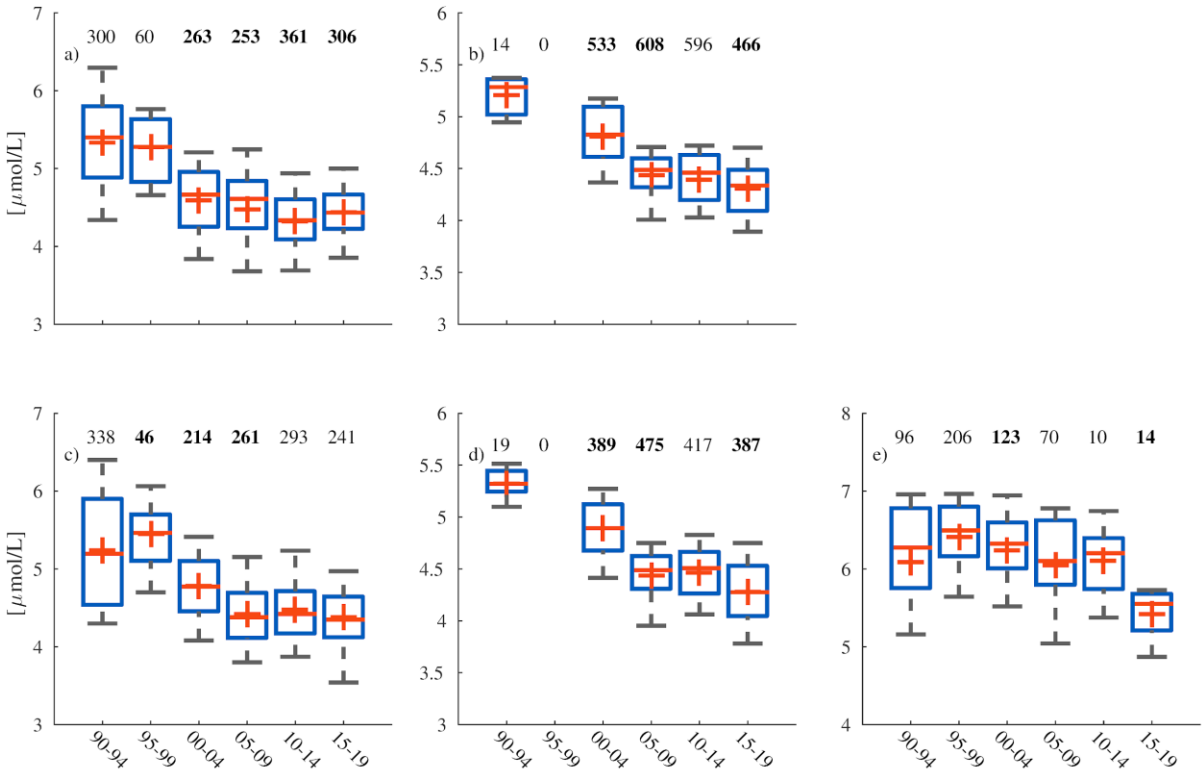


Figure 1.4.2. *Box-whisker plots of silicate concentrations (in micromol per litre; obtained from product ref. 1.4.1) bin-averaged in 5-year periods. Blue box shows 25th-75th percentile, red bar shows median, red cross shows mean, and the whiskers show min/max values. a) Norwegian Sea, 0-50 m. b) Barents Sea, 0-50 m. c) Norwegian Sea, 100-200 m. d) Barents Sea, 100-200 m. e) Greenland Sea, 100-200 m. Note the different scales on the y-axis. Sample size (n) is provided above each top whisker and shown in boldface where the change from the previous period is statistically significant (>95%).*

Initially, silicate in the Greenland Sea showed little or no decrease and showed no statistically significant decline between 1990-94 and 1995-99 (Fig. 1.4.2.). However, we observed a statistically significant decrease ($p < 0.05$) from 1995-1999 to 2000-2004, and the decline continued ($p < 0.05$) between the last two periods (2010-2014 and 2015-2019).

It is possible that silicate concentrations in the Greenland Sea not only reflect the conditions in Arctic waters, as this region can also be directly influenced by Atlantic water masses from adjacent regions and that this influence may change over the study period. Therefore, we used temperature as a proxy

for Atlantic Water influence as we compared silicate concentration and associated in-situ temperature. We omitted all observations where $T > 0$ °C, and our analysis revealed no correlation between silicate concentrations and in-situ temperature at each individual station (R = -0.02, p = 0.69; n = 519). We concluded that the observed changes in average silicate concentrations in the Greenland Sea were not caused by sampling in different water masses in different time-periods. Average temperature increased during the 5-year periods investigated (-0.86 °C, -0.85 °C, -0.60 °C, -0.40 °C, 0.41 °C and -0.25 °C, respectively). Average deeper layer silicate concentrations were overall, higher in the Greenland Sea for the entire period (6.0 µmol/L) compared to the Norwegian Sea (4.8 µmol/L) and the Barents Sea (4.5 µmol/L).

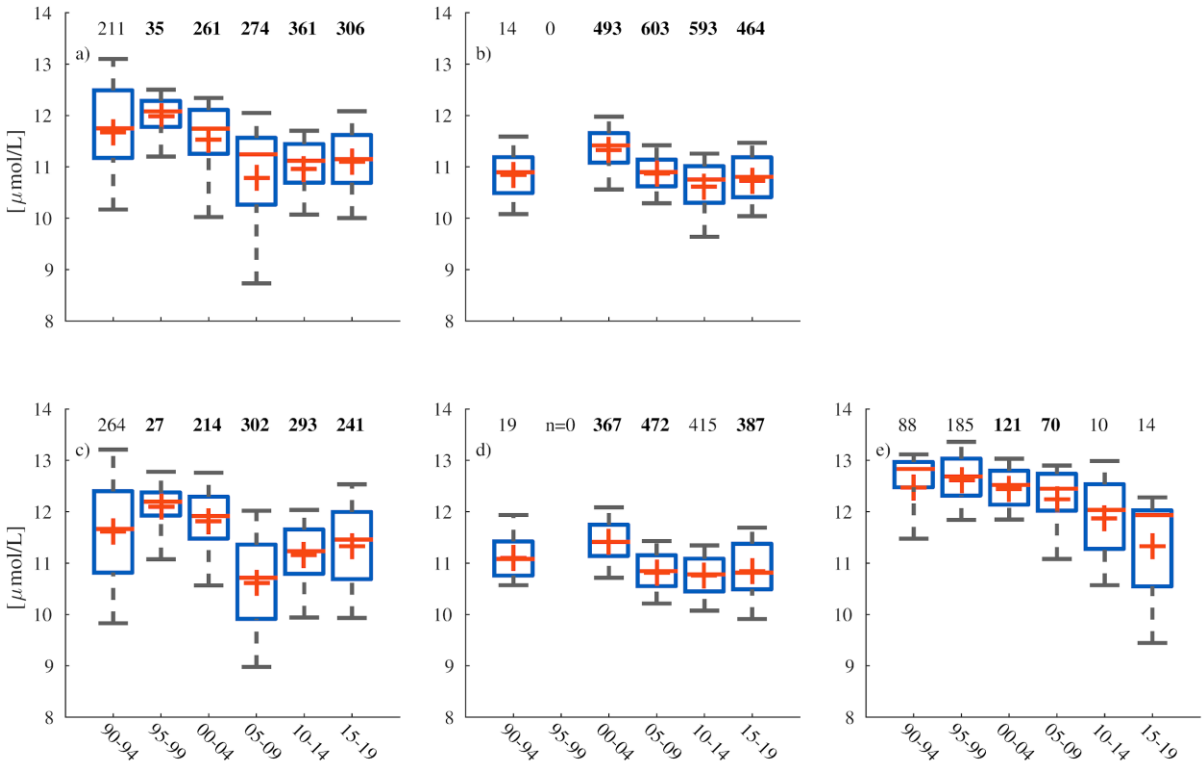


Figure 1.4.3. *Same as Figure 1.4.2, but for nitrate (obtained from product ref. 1.4.1).*

We observed a less dramatic decline in nitrate concentrations in all three regions (Fig. 1.4.3). The decline in nitrate appeared more pronounced (significant at the $p < 0.05$ level) in the Norwegian and Barents Seas (i.e., in AW) during the first half of the periods investigated, while the decrease in the Greenland Sea (i.e., in Arctic Water) appeared mostly in the latter half of the investigated period . In the Norwegian Sea, the average nitrate concentration decreased by 8 % in the euphotic zone and 9 % in the deeper waters, respectively, between 1990-1994 and 2005-2009. Subsequently, nitrate concentrations in the Norwegian Sea approached initial levels found in 1990-1994 (Fig. 1.4.3). In the Barents Sea, maximum nitrate concentrations were observed in 2000-2004, followed by a decrease of 6 % in both surface and deeper waters until 2010-14. The overall decline in nitrate concentrations in the Greenland Sea (9 %) was comparable to the other two regions in magnitude, but the drop was mostly confined to the latter half of the investigated period. Average concentrations of deep water nitrate were overall, higher in the Greenland Sea for the entire period (12.4 µmol/L) compared to the Norwegian Sea (11.3 µmol/L) and the Barents Sea (10.9 µmol/L).

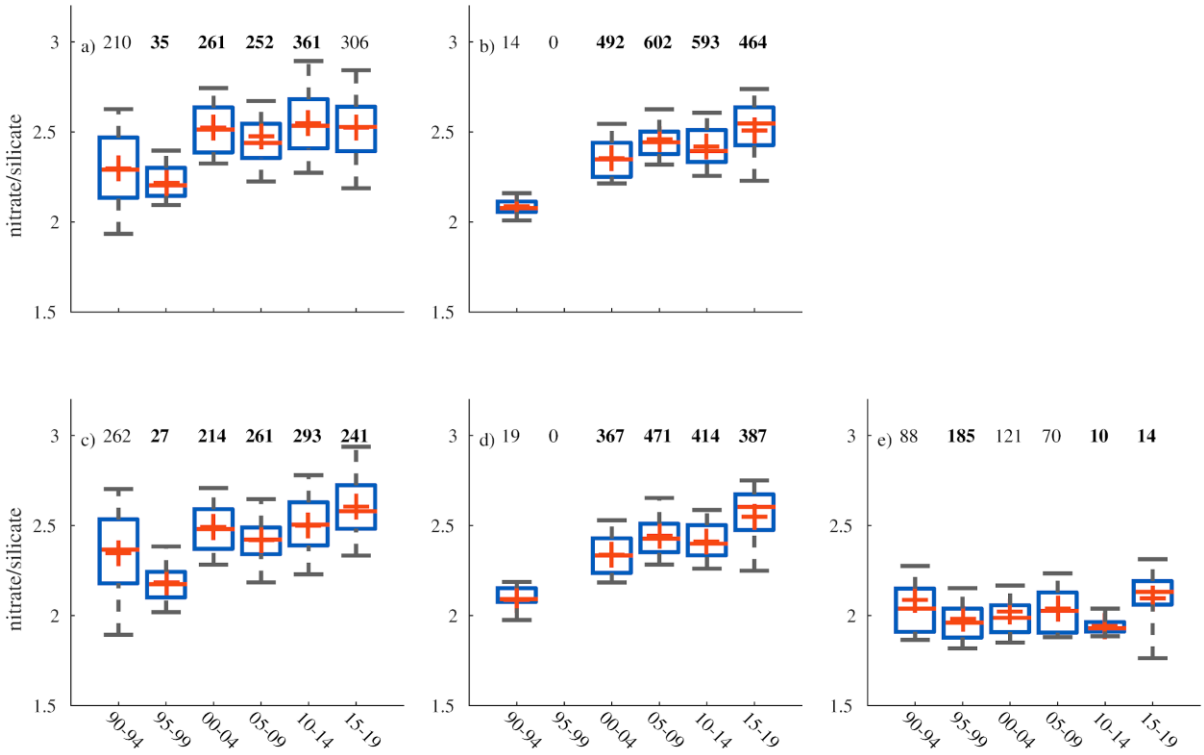


Figure 1.4.4. *Same as Figure 1.4.2, but for the nitrate:silicate-ratio (obtained from product ref. 1.4.1).*

While both silicate and nitrate concentrations showed an overall decline (Figs. 1.4.2 and 1.4.3), the relative proportion between the two macronutrients (the molar nitrate:silicate-ratio) showed an overall increase during the same period of time (Fig. 1.4.4). There was an initial decline to 2.2 in the Norwegian Sea (1995-1999) prior to an increase and a subsequent levelling off around 2.5, both in the euphotic zone and in deeper waters. In the Barents Sea the ratio was 2.1 in both the euphotic zone and the deeper layer in the first period (1990-94) before it increased to approximately 2.5. Note that we only have 19 and 14 data points (deep waters and surface layers, respectively) in the Barents Sea during the first 5-year period. Yet, the increase during this period (1990-1994) was significant at the 95 % confidence level. The observed nitrate:silicate-ratios were >2.5 during the last 5-year period in both the Norwegian and Barents Seas. In the Greenland Sea, the nitrate:silicate-ratio remained stable around 2 during the entire period investigated (Fig. 1.4.4).

1.4.4. Discussion

Decadal decline in silicate and nitrate, and the role of the SPG

Our compilation of deep water nutrient concentrations over the last 30 years (1990-2019) shows an abrupt and significant decline in silicate during the first decade (Fig. 1.4.2). The decline is slower for nitrate and reaches a minimum after 15-20 years, followed by a subsequent increase (Fig. 1.4.3). The decline in silicate is similar to previous bidecadal reports from the North Atlantic and subarctic regions (Rey et al. 2012, Hátún et al. 2017) but only one other study (Johnson et al. 2013) have reported a decline in nitrate (Rockall Trough, south of the Nordic Seas). Winter deep water mixing in open oceans will to a large extent determine nutrient concentrations in surface waters at the onset of the spring bloom

each year, and our data also shows that the reduction in silicate and nitrate is reflected in surface waters (0-50 m) in winter (Figs. 1.4.2 and 1.4.3). The largest decline appears during the first half of the period investigated, coinciding with a major weakening of the Subpolar Gyre Index (Fig. 1.4.5) as the index turned negative from 1994 to 1997. The SPG weakened further between 1999-2003, also with a subsequent impact on nutrient concentrations in the Norwegian and the Barents Seas (Fig. 1.4.2 and Fig. 1.4.3). We observed concurrent increases in nitrate concentrations during periods of minor strengthening of the SPG (2005-2009 and 2012-2015), while there was no uniform and significant reversals in our 5-year bin-averaging of deep water and surface silicate concentrations during these periods.

There is a statistically significant decline in Arctic deep water silicate in the Greenland Sea appearing approximately 15 years after the initial lowering of silicate and nitrate in Atlantic Water in the Norwegian and Barents Seas. Water entering the Greenland Sea can come from the North-Atlantic, the Arctic through the Fram Strait, or as a mixture of the two regions. The lack of correlation between silicate concentrations and sea temperature do suggest that the source water in the Greenland Sea gyre is of Arctic origin and entering the area through the Fram Strait. Also, since we observed a delay in decline of silicate in the Greenland Sea, relative to the adjacent Norwegian and Barent Seas (Fig. 1.4.2), we may rule out the North-Atlantic as a source. Water travelling from the Norwegian and Barents Seas, through the Polar Basin and the East Greenland Current, to the Greenland Sea, has an approximate periodicity of several years to a few decades, depending on the depth (Schlosser et al. 1995). Therefore, the decadal-scale time lag of the decrease in nutrients in the Greenland Sea (Fig. 1.4.2) suggests that nutrient concentrations in Arctic waters (indirectly spurred by changes in the Norwegian and Barents Seas) are also determining bioavailable silicate and nitrate in the Greenland Sea.

The decline in biologically available silicate throughout the northern North Atlantic in surface waters, has been attributed to changes in the SPG circulation (Hátún et al. 2017). This appears to concur with the decline in silicate, and nitrate, concentrations reported from the Norwegian and Barents Seas in this study, and is similar in scope to the observations made by Johnson et al. (2013) for nitrate and phosphate at the southern entrance to the Nordic Seas, in the Rockall Trough area. The drop in nutrient concentrations have already been attributed to a weakening and westward movement of the Subpolar Gyre Index (Rey 2012, Johnsen et al. 2013, Hatun et al. 2017) as less nutrient rich North Atlantic water has entered the Nordic Seas. Therefore, our reported decline in both silicate and nitrate is similar to other findings in the same region and concur with an almost unidirectional weakening of the Subpolar Gyre Index in the period investigated (Fig. 1.4.5).

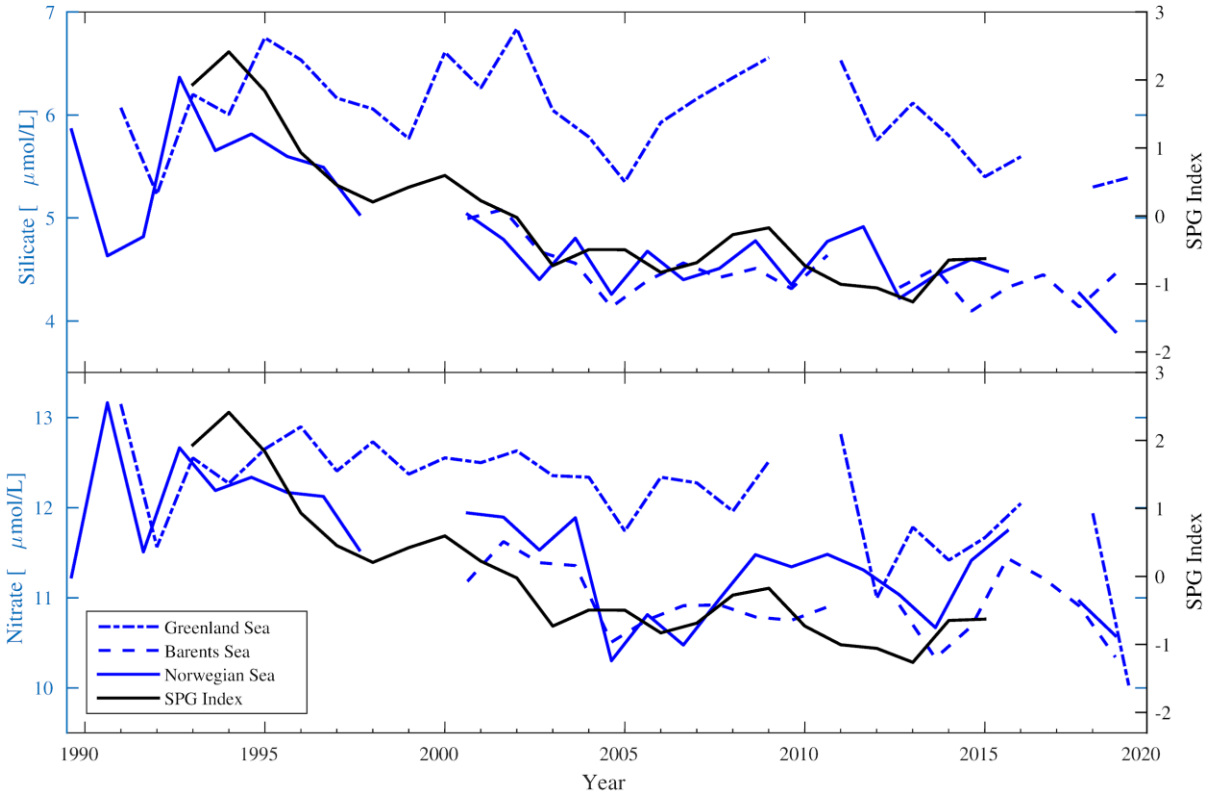


Figure 1.4.5. *Time series of silicate concentrations (top panel, blue lines) and nitrate concentrations (bottom panel, blue lines) in the Norwegian Sea (solid line), the Barents Sea (dashed line) and the Greenland Sea (dash-dotted line). The data were obtained from product ref. 1.4.1. Black lines show the annual Subpolar Gyre Index (data from Birx & Paye 2016). Values for the Norwegian and Barents seas represent January-March averages, and values from the Greenland Sea are annual deep water averages.*

Phytoplankton community composition and zooplankton grazing

During an annual growth season, Officer & Ryther (1980) suggested two basic phytoplankton communities; the diatom-dominated ones and flagellated (non-diatom) primary producers. Due to their rapid growth at high nutrient concentrations (the spin-up for spring blooms), the former community is outcompeting other phytoplankton at silicate concentrations > 2 µM, whereas flagellated cells only flourish at lower nutrient concentrations and with silicate concentration < 2 µM (Furnas 1990, Egge & Aksnes 1992). The spring diatom bloom in the Nordic Seas is often followed by flagellated, non-diatom phytoplankton, only interrupted by smaller episodes of minor diatom blooms towards the end of the growth season in the fall (McQuatters-Gollop et al. 2007). Zooplankton growth and development is closely tied in with feeding efficiency (Pond et al. 1996) and measured clearance and ingestion rates for *Calanus finmarchicus* are at a peak during diatom spring blooms in the Norwegian Sea (Meyer-Harms et al. 1999). Growth development and reproduction of zooplankton, such as *C. finmarchicus* in the Norwegian and Barents Seas, is highly dependent of the timing and magnitude of the spring diatom blooms in these oceans (Melle et al. 2004, Eiane & Tande 2009). Juvenile and to some extent adult fish, such as the commercial stocks of herring, cod and capelin in the Norwegian and the Barents Seas, are in turn strongly dependent on successful zooplankton growth and reproduction each year (Skjoldal et al. 2004, Gjøsæther et al. 2009). Therefore, lowered silicate concentrations may lead to less extensive

spring diatom blooms and hence, less successful growth and survival of zooplankton and, eventually, fish larvae from commercial stocks in these oceans.

Chemosensory detection of phytoplankton means that zooplankton can discriminate between toxic and non-toxic cells, even within the same species expressing different levels of toxicity (e.g. Selander et al. 2006). The chemosensory capabilities in zooplankton are also used to assess food particle quality in order to optimize the intake of high quality foods (Mayzaud et al. 1996, Olsen et al. 2006). We still do not have a complete understanding of all the ques that determine "high quality foods" but copepods are also known to be able to discriminate different growth phases (and hence, a potentially different nutritional quality) in the same species of phytoplankton (Barofsky et al. 2010). This change in the "phycosphere" (Moore et al. 1999) surrounding each phytoplankton cell will, amongst others, depend on nutrient availability and associated physical/chemical stress factors, that may create different reproductive outcomes and growth in zooplankton. Changes in the relative proportion of available nutrients (e.g. the nitrate:silicate-ratio) may also create stress in phytoplankton (diatoms) who are depending on silicate for cellular growth. Calculated nitrate:silicate-ratios in this study (Fig 1.4.3) were well above a molar ratio of 2 and hence, proportionately more nitrate than silicate was available at the onset of the spring bloom in all three regions investigated. This may indicate however, that if the decline in silicate are continuing, and on a larger scale than e.g. the decline in nitrate, the prerequisites for successful diatom spring blooms (abundance of silicate) may no longer become significant. Successful reproductive outcomes for zooplankton (egg production and survival of nauplii) is highly dependent on access to an abundance of diatoms of optimal nutritional value (Jonasdottir 1994, Irigoien et al. 2000). Therefore, limited access to silicate for diatoms growth and development, may hamper zooplankton egg production and survival of zooplankton nauplii. If the negative Subpolar Gyre Index persist, the Nordic Seas may see longer periods of surface waters with silicate levels below 2 µM (critical for diatom growth) and reduced growth and recruitment in zooplankton (mainly *Calanus* sp.) may in turn have consequences for growth and development of fish larvae of commercially important stocks in these waters

Product used

Ref. No.	Product name & type	Documentation
1.4.1	INSITU_GLO_BGC_REP_OBSERVATIONS_013_046	**PUM:** http://marine.copernicus.eu/documents/PUM/CMEMS-INS-PUM-013-046.pdf **QUID:** http://marine.copernicus.eu/documents/QUID/CMEMS-INS-QUID-013-046.pdf

REFERENCES

Section 1.4

Barofsky, A., et al. (2010) Growth phase of the diatom *Skeletonema marinori* influences the metabolic profile of the cells and the selective feeding of the copepod *Calanus* spp. J. Plankt. Res. 32:263-272

Bendschneider, K. & R.I. Robinson (1952) A new spectrophotometric method for the determination of nitrite in seawater. J. Mar. Res. 2:87-96

Berx, B. & M. Payne (2016) Sub-Polar Gyre Index, doi: 10.7489/1806-1

Egge, J. & D. Aksnes (1992) Silicate as regulating nutrient in phytoplankton competition. Mar. Ecol. Prog. Ser. 83:281-289

Eiane, K. & K.S. Tande (2009) Meso and microzooplankton, pp.209-234. In: Sakshaug, E., G. Johnsen, K. Kovac (Eds), Ecosystem Barents Sea, 587 pp.

Furnas, M.J. (1990) In situ growth rates of marine phytoplankton: approaches to measurement, community and species growth rate. J. Plankton Res. 12:1117-1151

Gjøsæther, H. (2009) Commercial fisheries (fish, seafood and marine mammals), pp.373-414. In: Sakshaug, E., G. Johnsen, K. Kovac (Eds), Ecosystem Barents Sea, 587 pp.

Grasshoff, K. (1965) On the Automatic Determination of Phosphate, Silicate and Fluoride in Seawater. ICES Hydrographic Committee Report No. 129

Hátún, H. et al. (2005) Influence of the Atlantic Subpolar Gyre on the Thermohaline Circulation. Science 309:1841-1844

Hátún, H. et al. (2017) The subpolar gyre regulates silicate concentrations in the North Atlantic. Sci. Rep. 7, 14576

Honjo, S. & S.J. Manganini (1993) Annual biogenic particle fluxes to the interior of the North Atlantic Ocean studied at 34-degrees N 21-degrees W and 48 degrees N 21 degrees W. Deep-Sea Res. II 40:587-607

Irigoien, X., et al. (2000) Feeding selectivity and egg production of *Calanus helgolandicus* in the English Channel. Limnol. Oceanogr. 45:44-54

Jaccard, P. et al. (2018) Quality Control of Biogeochemical Measurements, INSITU TAC CMEMS ELEMENT, 47 pp, http://doi.org/10.13155/36232

Jaccard, P. et al. (2020) Quality Information Document for global ocean reprocessed in-situ observations of biogeochemical products. Issue 2.0, 77 pp. http://doi.org/10.13155/54846

Johnson, C., et al. (2013) Declining nutrient concentrations in the northeast Atlantic, as a result of a weakening Subpolar Gyre. Deep-Sea Res. 82:95-107

Jonasdottir, S.H. (1994) Effects of food quality on reproductive success in *Acartia tonsa* and *Acartia hudsonica* - laboratory observations. Mar. Biol. 121:67-81

Mayzaud, P., et al. (1996) The influence of food quality on the nutritional acclimation of the copepod *Acartia clausi*. 6th International Conference on Copepoda., Elsevier, Germany, pp.483-493

McQuatters-Gollop, A., et al. (2007) A long-term chlorophyll data set reveals regime shift in North Sea phytoplankton biomass unconnected to nutrient trends. Limnol. Oceanogr. 52:635–648

Melle, W., et al. (2004) Zooplankton: the link to higher trophic levels. In: Skjoldal, H.R. (Ed.) The Norwegian Sea Ecosystem, pp.209-234, Tapir, Norway

Meyer-Harms, B., et al. (1999) Selective feeding on natural phytoplankton by *Calanus finmarchicus* before, during, and after the 1997 spring bloom in the Norwegian Sea. Limnol. Oceanogr. 44:154-165

Moore, P.E., et al. (1999) Physical constraints of chemoreception in foraging copepods. Limnol. Oceanogr. 44:166-177

Officer, C.B. & J.H. Ryther (1980) The possible importance of silicon in marine eutrophication. Mar. Ecol. Prog. Ser. 3:83-91.

Olsen, M.B., et al. (2006) Copepod feeding selectivity on microplankton, including the toxigenic diatoms, *Pseudo-nitzchia* spp. in the coastal Pacific Northwest. Mar. Ecol. Prog. Ser. 326:207-220

Pollock, D.E. (1997) The role of diatoms, dissolved silicate and Antarctic glaciation in glacial/interglacial climate change; a hypothesis. Global Planet Change 14: 113-125.

Pond, D., et al. (1996) Environmental and nutritional factors determining seasonal variability in the fecundity and egg viability of *Calanus helgolandicus* in coastal waters off Plymouth, UK. Mar. Ecol. Prog. Ser. 143:45-63

Rey, F. (2004) Phytoplankton: the grass of the ocean. In: Skjoldal, H.R. (Ed.) The Norwegian Sea Ecosystem, pp.97-136, Tapir, Norway

Rey, F. (2012) Declining silicate concentrations in the Barents Sea. ICES J. Mar. Sci. 69:208-212

Sarafanov, A. (2009) On the effect of the North Atlantic Oscillation on temperature and salinity of the subpolar North Atlantic intermediate and deep waters. ICES J. Mar. Sci. 66:1448-1454

Savidge, G., et al. (1995) A study of the spring bloom in the N-E Atlantic Ocean in 1990. Deep Sea Res. 42:599-617

Schlosser, P., et al. (1995) The role of the large-scale Arctic Ocean circulation in the transport of contaminants. Deep-Sea Research II, 42:1341-1367

Selander, E., et al. (2006) Copepods induce paralytic shellfish toxin production in marine dinoflagellates. Proc. R. Soc. Lond. Ser. B, 273:1673-1680

Skjoldal, H.R., et al. (2004) Food webs and trophic interactions. In: Skjoldal, H.R. (Ed.) The Norwegian Sea Ecosystem, pp.447-506, Tapir, Norway

Smayda, T.J. (1990) Novel and nuisance phytoplankton blooms in the sea: evidence for global epidemic. In: Graneli, E., et al. (eds) Toxic marine phytoplankton. Elsevier, NY, p29-40

Volk, T. & M.I. Hoffert (1985) Ocean Carbon Pumps: Analysis of Relative Strengths and Efficiencies in Ocean-Driven Atmospheric CO_2 Changes. In: Sundquist, E.T. & W.S. Broecker (Eds) The Carbon Cycle and Atmospheric CO2: Natural Variations Archean to Present, Geophysical Monograph Series, Vol. 32, https://doi.org/10.1029/GM032p0099